

Solution 1

Question 1 (a)

Experiments with the EM algorithm: In this problem, we'll generate data from a mixture of Gaussians and then see whether the EM algorithm is able to recover the correct clustering and the correct parameters of the mixture model.

Consider a mixture of four Gaussians in $\mathbb{R}^2$ such that:

$$p(x) = \pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \pi_3 N(\mu_3, \Sigma_3) + \pi_4 N(\mu_4, \Sigma_4)$$

where the mixing weights $\pi_i \ni i \in \{1, 2, 3, 4\}$ and $\pi_i \in \mathbb{R}$ are given by:

$$\pi_1 = 0.1, \pi_2 = 0.2, \pi_3 = 0.5, \pi_4 = 0.2$$

the cluster means $\mu_i \in \mathbb{R}^2$ are given by:

$$\mu_1 = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \mu_4 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

and the covariance matrices $\Sigma_i \in \mathbb{R}^{2 \times 2}$ are given by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 25 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 9 & 1 \\ 1 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} 16 & -6 \\ -6 & 4 \end{bmatrix}$$

- Generate 400 points at random from this mixture model.
- Plot the points, using colors to distinguish the four components. Put this plot in your solution.

Solution 1 (a)

Python Code

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # set random seed
5 np.random(42)
6
7 # define mixture model parameters
8
9 # generate 400 random points from the mixture model
10
11 # plot points, using colors to distinguish each of the components

```

Plot

∴

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Solution 1

Question 1 (b)

Experiments with the EM algorithm: In this problem, we'll generate data from a mixture of Gaussians and then see whether the EM algorithm is able to recover the correct clustering and the correct parameters of the mixture model.

Consider a mixture of four Gaussians in $\mathbb{R}^2$ such that:

$$p(x) = \pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \pi_3 N(\mu_3, \Sigma_3) + \pi_4 N(\mu_4, \Sigma_4)$$

where the mixing weights $\pi_i \ni i \in \{1, 2, 3, 4\}$ and $\pi_i \in \mathbb{R}$ are given by:

$$\pi_1 = 0.1, \pi_2 = 0.2, \pi_3 = 0.5, \pi_4 = 0.2$$

the cluster means $\mu_i \in \mathbb{R}^2$ are given by:

$$\mu_1 = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \mu_4 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

and the covariance matrices $\Sigma_i \in \mathbb{R}^{2 \times 2}$ are given by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 25 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 9 & 1 \\ 1 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} 16 & -6 \\ -6 & 4 \end{bmatrix}$$

- Write a routine that takes a two-dimensional Gaussian mixture (means and covariance matrices) and plots ellipses corresponding to each component. Using this routine, plot all the data points (from (a)) in the same color and then draw four ellipses on the same plot, depicting representative contours of the four Gaussians.

Solution 1 (b)

Python Code

NOTE: points from part(a) should be on same plot as ellipses

```

1  # from part(a)
2  import numpy as np
3  import matplotlib.pyplot as plt
4
5  # set random seed
6  np.random(42)
7  # define mixture model parameters
8
9  # generate 400 random points from the mixture model
10
11 # plot points, using colors to distinguish each of the components
12
13
14 # routine that takes 2D Gaussian mixture(mu_i, sigma_i) and plots ellipses corresponding
15   to each component
16 from matplotlib.patches import Ellipse
17 eigvals, eigvecs = np.linalg.eigh(sigma)
18 direction = eigvecs[0] / np.linalg.norm(eigvecs[0])
19 theta = np.arctan2(direction[1], direction[0]) * 180 / np.pi + 180
20 stretch = 2.0 * np.sqrt(3.0 * eigvals)
21 ellipse = Ellipse(mu, stretch[0], stretch[1], angle=theta, color='red')
22 ellipse.set_alpha(0.25)
23 ax.add_artist(ellipse)

```

Plot

∴

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Solution 1

Question 1 (c)

Experiments with the EM algorithm: In this problem, we'll generate data from a mixture of Gaussians and then see whether the EM algorithm is able to recover the correct clustering and the correct parameters of the mixture model.

Consider a mixture of four Gaussians in $\mathbb{R}^2$ such that:

$$p(x) = \pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \pi_3 N(\mu_3, \Sigma_3) + \pi_4 N(\mu_4, \Sigma_4)$$

where the mixing weights $\pi_i \ni i \in \{1, 2, 3, 4\}$ and $\pi_i \in \mathbb{R}$ are given by:

$$\pi_1 = 0.1, \pi_2 = 0.2, \pi_3 = 0.5, \pi_4 = 0.2$$

the cluster means $\mu_i \in \mathbb{R}^2$ are given by:

$$\mu_1 = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \mu_4 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

and the covariance matrices $\Sigma_i \in \mathbb{R}^{2 \times 2}$ are given by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 25 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 9 & 1 \\ 1 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} 16 & -6 \\ -6 & 4 \end{bmatrix}$$

- Look at the plots from (a) and (b) and check that the shapes of the clusters correspond to what you would expect from the covariance matrices. One of the clusters is small and is partially contained within another cluster. Which clusters are these?

Solution 1 (c)

Plot

Plot Comparison Comments

One of the clusters is small and is partially contained within another cluster. Which clusters are these?

$\therefore$

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Solution 1

Question 1 (d)

Experiments with the EM algorithm: In this problem, we'll generate data from a mixture of Gaussians and then see whether the EM algorithm is able to recover the correct clustering and the correct parameters of the mixture model.

Consider a mixture of four Gaussians in $\mathbb{R}^2$ such that:

$$p(x) = \pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \pi_3 N(\mu_3, \Sigma_3) + \pi_4 N(\mu_4, \Sigma_4)$$

where the mixing weights $\pi_i \ni i \in \{1, 2, 3, 4\}$ and $\pi_i \in \mathbb{R}$ are given by:

$$\pi_1 = 0.1, \pi_2 = 0.2, \pi_3 = 0.5, \pi_4 = 0.2$$

the cluster means $\mu_i \in \mathbb{R}^2$ are given by:

$$\mu_1 = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \mu_4 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

and the covariance matrices $\Sigma_i \in \mathbb{R}^{2 \times 2}$ are given by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 25 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 9 & 1 \\ 1 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} 16 & -6 \\ -6 & 4 \end{bmatrix}$$

- Now run the EM algorithm on the data. Plot the data, and using the procedure from (b), draw four ellipses on the same plot, depicting the four Gaussians returned by EM. How close are they to the correct Gaussians?

Solution 1 (d)

Python Code

```
1 from sklearn.mixture import GaussianMixture
```

Plot

Plot Comparision Comments

How close are they to the correct Gaussians?

∴

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Solution 1

Question 1 (e)

Experiments with the EM algorithm: In this problem, we'll generate data from a mixture of Gaussians and then see whether the EM algorithm is able to recover the correct clustering and the correct parameters of the mixture model.

Consider a mixture of four Gaussians in $\mathbb{R}^2$ such that:

$$p(x) = \pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \pi_3 N(\mu_3, \Sigma_3) + \pi_4 N(\mu_4, \Sigma_4)$$

where the mixing weights $\pi_i \ni i \in \{1, 2, 3, 4\}$ and $\pi_i \in \mathbb{R}$ are given by:

$$\pi_1 = 0.1, \pi_2 = 0.2, \pi_3 = 0.5, \pi_4 = 0.2$$

the cluster means $\mu_i \in \mathbb{R}^2$ are given by:

$$\mu_1 = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \mu_4 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

and the covariance matrices $\Sigma_i \in \mathbb{R}^{2 \times 2}$ are given by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 25 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 9 & 1 \\ 1 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} 16 & -6 \\ -6 & 4 \end{bmatrix}$$

- Plot the data again, but this time color each point according to it most likely mixture component. How does this clustering compare to the true component memberships from *part (a)*? Are dependencies understandable?

Solution 1 (e)

Python Code

```
1 from sklearn.mixture import GaussianMixture
```

Plot

Plot Comments

How does this clustering compare to the true component memberships (from (a))?

Are the discrepancies understandable?

∴

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Solution 1

Question 1 (f)

Experiments with the EM algorithm: In this problem, we'll generate data from a mixture of Gaussians and then see whether the EM algorithm is able to recover the correct clustering and the correct parameters of the mixture model.

Consider a mixture of four Gaussians in $\mathbb{R}^2$ such that:

$$p(x) = \pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \pi_3 N(\mu_3, \Sigma_3) + \pi_4 N(\mu_4, \Sigma_4)$$

where the mixing weights $\pi_i \ni i \in \{1, 2, 3, 4\}$ and $\pi_i \in \mathbb{R}$ are given by:

$$\pi_1 = 0.1, \pi_2 = 0.2, \pi_3 = 0.5, \pi_4 = 0.2$$

the cluster means $\mu_i \in \mathbb{R}^2$ are given by:

$$\mu_1 = \begin{bmatrix} 7 \\ 6 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \mu_4 = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

and the covariance matrices $\Sigma_i \in \mathbb{R}^{2 \times 2}$ are given by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 25 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 9 & 1 \\ 1 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} 16 & -6 \\ -6 & 4 \end{bmatrix}$$

- Finally, let's see how EM's solution evolves over iterations of local search. Show four plots, depicting the progress of EM at different iterations.

Solution 1 (f)

Python Code

```
1 from sklearn.mixture import GaussianMixture
```

Plot

∴

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Solution 2

Question 2 (a)

Kernel density estimation: Suppose we have samples $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ from an unknown distribution f , and we want to come up with a model of f . One simple way to do this is using the kernel density estimator $\hat{f}$, which approximates f using a mixture of n Gaussians, each with weight $1/n$, and each centered at one of the data points:

$$\hat{f} : \frac{1}{n}N(x_1, \sigma^2 I_d) + \frac{1}{n}N(x_2, \sigma^2 I_d) + \dots + \frac{1}{n}N(x_n, \sigma^2 I_d)$$

Here σ must be selected carefully. It is usually called the bandwidth parameter.

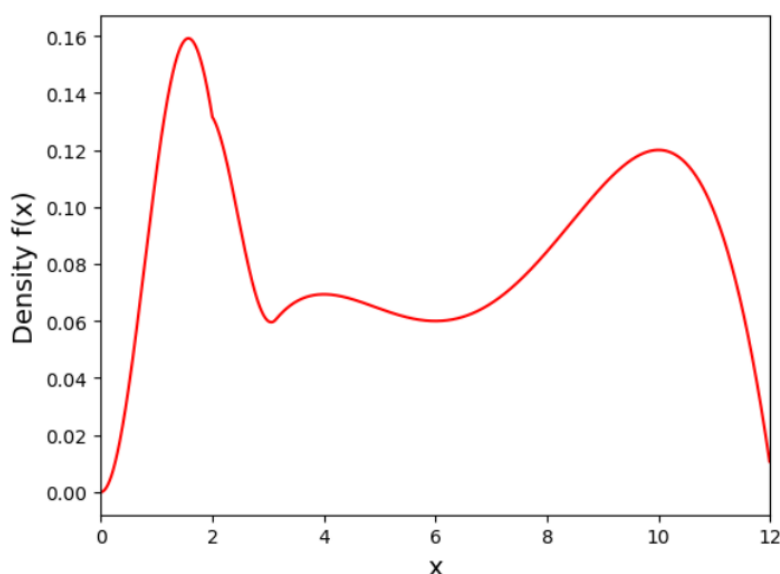


Figure 1: The file `samples.txt` contains 1,000 samples drawn from f (using rejection sampling).

- Plot a histogram of the 1,000 samples using 20 bins. It should (very roughly) resemble the distribution f shown above.

Solution 2 (a)

Python Code

```
1 import matplotlib.pyplot as plt
```

Plot

∴

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Solution 2

Question 2 (b)

Kernel density estimation: Suppose we have samples $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ from an unknown distribution f , and we want to come up with a model of f . One simple way to do this is using the kernel density estimator $\hat{f}$, which approximates f using a mixture of n Gaussians, each with weight $1/n$, and each centered at one of the data points:

$$\hat{f} : \frac{1}{n}N(x_1, \sigma^2 I_d) + \frac{1}{n}N(x_2, \sigma^2 I_d) + \dots + \frac{1}{n}N(x_n, \sigma^2 I_d)$$

Here σ must be selected carefully. It is usually called the bandwidth parameter.

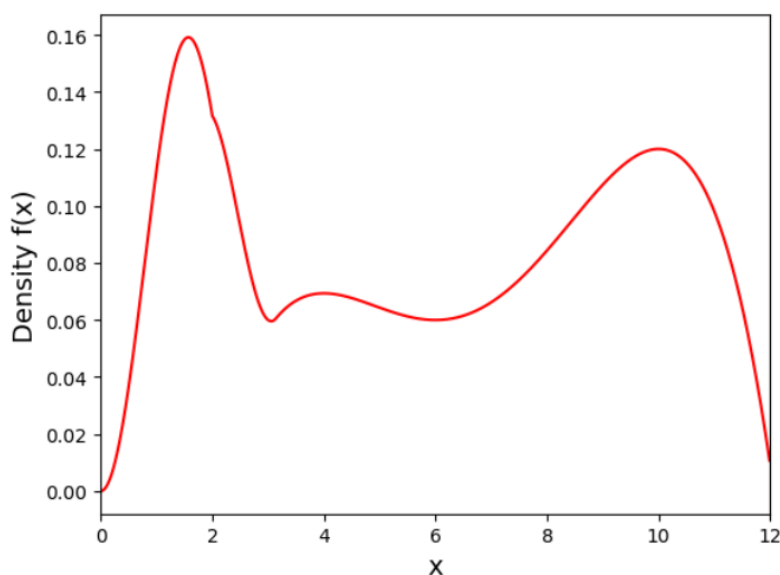


Figure 2: The file `samples.txt` contains 1,000 samples drawn from f (using rejection sampling).

- Now apply kernel density estimation using just the first 100 samples in `samples.txt`, that is, $n = 100$. For the bandwidth parameter, try the following settings: 0.25, 0.5, 0.75, 1.0.
 - For each of the four bandwidth settings, show a plot of the resulting density $\hat{f}$.
 - What changes as the bandwidth increases? Give a one- or two-sentence description.
 - In this case, which of the four bandwidth settings yields a model $\hat{f}$ that is closest to the original f , in your opinion?

Solution 2 (b)

Python Code

```

1 from sklearn.neighbors import KernelDensity
2 from matplotlib.pyplot as plt
3 from scipy.stats import norm
4 import numpy as np
5
6 # define density settings
7 densities = [0.25, 0.50, 0.75, 1.0]
```

Plot**Plot Comments**

What changes as the bandwidth increases? Which bandwidth setting yeilds a model $\hat{f}$ that is closest to the original f , in your opinion?

$\therefore$

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Solution 2

Question 2 (c)

Kernel density estimation: Suppose we have samples $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ from an unknown distribution f , and we want to come up with a model of f . One simple way to do this is using the kernel density estimator $\hat{f}$, which approximates f using a mixture of n Gaussians, each with weight $1/n$, and each centered at one of the data points:

$$\hat{f} : \frac{1}{n}N(x_1, \sigma^2 I_d) + \frac{1}{n}N(x_2, \sigma^2 I_d) + \dots + \frac{1}{n}N(x_n, \sigma^2 I_d)$$

Here σ must be selected carefully. It is usually called the bandwidth parameter.

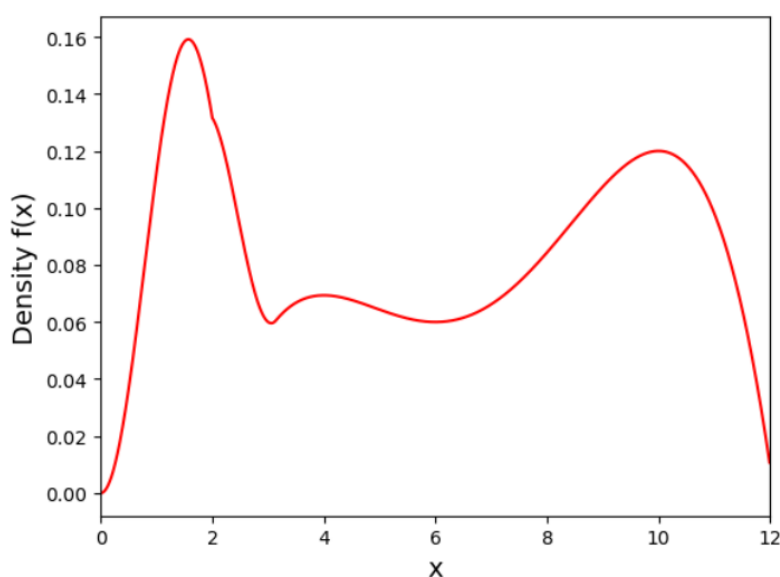


Figure 3: The file `samples.txt` contains 1,000 samples drawn from f (using rejection sampling).

- Now show the plots of the kernel density estimate using all 1000 samples in `samples.txt`, that is, $n = 1000$. Use the same four settings of the bandwidth.

Solution 2 (c)

Python Code

```

1 from sklearn.neighbors import KernelDensity
2 from matplotlib.pyplot as plt
3 from scipy.stats import norm
4 import numpy as np
5
6 # define density settings
7 densities = [0.25, 0.50, 0.75, 1.0]
```

Plot

∴

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